There are 3 tasks given in the document. You may do ANY ONE of your choice

Task 1 - LLM Sandbox

You are one of the developers working on the backend for a TechnoVIT event by GDG VIT Chennai, where participants are expected to "break" the sandboxed systems to get access and hints to the next round. You have been assigned the task to develop the system for one of the rounds. The task of this round is prompt injection, where the solution to this round is embedded in the system prompt.

Develop a backend system that accepts a user prompt, processes it through an LLM (self-hosted, third party or participant-given credentials), and proceeds to give the result. You have to ensure the system prompt is **just resilient enough**, and the backend itself should be feasible, scalable and cost-efficient to deploy.

Task 2 - Remote Runtime Environment

You are leading the backend team for GDG VIT Chennai's speed-coding event. In this event, participants compete against each other with algorithmic efficiency and problem solving.

Develop a Remote Runtime Environment (multi-language support is optional), where the code given by participants would be compiled, run in the environment, and efficiency is matched. The system needs to be deterministic, and also be able to handle disruptions in the environment like process preemption, memory leaks, infinite loops, memory overflow, etc.

Task 3 - Event Management Backend

Note: This task is for FRESHERS ONLY

As the sole developer responsible for creating the backend for GDG VIT Chennai's flagship event (because the seniors are too busy with placements), you have been tasked with developing the backend system that connects organisers, judges and participants.

Develop the backend for the app/website, with suitable RBAC and a core judging workflow, along with a realtime chat for the organisers and participants. You must also address common security threats such as CSRF. Hundreds of participants attend this event across the country, hence your system should be able to cope up without incurring too much cost.

Location tracking, remote alerts, and other features may be implemented as extensions.

Submission

Deadline: 29th August (Saturday), 23:59:59

1. Create a `README.md` file detailing how the project works, the problems encountered and the mitigation steps. It should also describe how edge cases, access control, and other parts are handled
2. Upload the project code to github in a public repository
3. Take a video demonstrating the working of the project. The video should not be more than 5 minutes long. Upload it to Google Drive with the necessary permissions.
4. Deploy the project on a platform of your choice (Render, Vercel, AWS, Firebase, etc.)
5. Fill the form: https://forms.gle/ckVwxiGmYvg4J4ZN7

Evaluation Criteria

> ⚠ **Attention**
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> - The project need not be flawless.
> - Prioritize a well-designed working core over feature completeness.
> - AI assistance is allowed, but the implementation and decisions should primarily be your own, and you should understand everything you submit.

Criteria	Description	Points
Documentation	Working, data flow, security, tech stack and any supporting diagrams.	20
Code Quality	The code should be extensible, follow best principles, and easily readable.	15
Scalability and Cost Efficiency	The system should be scalable. This includes database optimization, system design and code architecture	25
Reproducibility and Development	The system should be easily reproducible on the cloud environment, and also be equally easy to modify	25
Tech Stack	How efficient and relevant the tech stack is for the given scenario, including database and framework	15